

# Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The paper is silent on annotator demographics, qualifications, IAA protocols, and label-validation procedures — Output Content concerns are explicitly NOT DOCUMENTED in the benchmark write-up. Questions were assembled by US graduate/undergraduate students [\[Q17\]](#) and validated against US-standard answer keys. No Indian SMEs, UPSC/SSC specialists, or Hindi-medium educators participated. For India-relevant items that do appear ([DATASET-D15](#) 1857 uprising), the answer keys privilege colonial historiography over NCERT/Bipin Chandra nationalist framing, directly conflicting with what Indian exam stakeholders would judge correct. moral\_scenarios labels are explicitly anchored to 'ordinary moral standards in the US as of 2020' [\[DATASET-D18, D19\]](#), producing systematic OC misalignment for the target population. Even the strongest Indian-grounded benchmarks (ParamBench, MILU) only partially document Indian-SME involvement [\[WEB-15, WEB-13\]](#); MMLU documents none.

| ID                     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q17</a>    | "The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3) |
| <a href="#">WEB-15</a> | ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks ( <a href="https://arxiv.org/abs/2508.16185">https://arxiv.org/abs/2508.16185</a> )                                                                                                                                                                                                                  |
| <a href="#">WEB-13</a> | MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks ( <a href="https://aclanthology.org/2025.naacl-long.507/">https://aclanthology.org/2025.naacl-long.507/</a> )                                                                                                                                                                                      |
| <a href="#">D15</a>    | the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing                                                       |
| <a href="#">D18</a>    | "according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."                                                                                                                                                                                                                                                               |
| <a href="#">D19</a>    | "according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."                                                                                                                                                                                                                                                                   |

## Strengths

- For purely-cultural-neutral STEM items (abstract algebra, conceptual physics), label correctness is unlikely to depend on annotator culture — these labels probably transfer to Indian stakeholders [\[DATASET-D27, D30, D40\]](#).
- Memorization analysis suggests low risk that exact questions were leaked into pretraining [\[Q98, Q99\]](#), so labels are not artifacts of test contamination.

| ID                  | Evidence                                                                                                                                                                                                                            |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">D27</a> | Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context                   |
| <a href="#">D30</a> | Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$ . — Abstract mathematics — entirely culture-independent                                                                                                      |
| <a href="#">D40</a> | A wave transfers ... energy — Basic physics principle — universally applicable                                                                                                                                                      |
| <a href="#">Q98</a> | "However, in Figure 13, we see that accuracy and question entropy are not positively correlated, suggesting that the test's low-entropy questions do not correspond to memorized (and thereby correctly predicted) answers." (p.14) |

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|-----|---------------------------------------------------------------------|
| Q99 | "This suggests that our exact questions were not memorized." (p.14) |
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## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                               |
|------|-------------------------------------------------------------------|
| OC-1 | Do ground-truth labels reflect regional stakeholder perspectives? |
| OC-2 | Assess potential annotator-regional population disagreement       |
| OC-3 | Review annotator demographics from documentation                  |
| OC-4 | Consider label re-annotation by regional annotators               |
| OC-5 | Review aggregation methods for minority perspective erasure       |
| OC-6 | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** No — labels reflect US-educated annotator perspectives [Q17]; for India-centric content, Indian stakeholder perspectives are unrepresented.

**OC-2:** Significant disagreement is plausible for moral\_scenarios (US 2020 norms vs. Indian norms) [DATASET-D18, D19], for high\_school\_world\_history items framed via colonial sources [DATASET-D15, D47], and for any Indian-content item validated against non-Indian answer keys.

**OC-3:** INSUFFICIENT DOCUMENTATION — paper does not include a Datasheet or Data Statement section addressing annotator demographics; only [Q17] notes 'graduate and undergraduate students' without further detail.

**OC-4:** Re-annotation by representative Indian annotator pool is necessary for any India-deployment use; ParamBench [WEB-15] provides a partial template but does not cover MMLU items.

**OC-5:** INSUFFICIENT DOCUMENTATION — aggregation methods for label disagreements are not described; cannot assess minority-perspective erasure.

**OC-6:** Convergent and external validity are severely harmed: MMLU labels do not correlate with Indian-stakeholder judgments on culturally embedded items, and US-norm-anchored labels do not generalize to Indian deployment context.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q17    | "The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3) |
| D18    | "according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."                                                                                                                                                                                                                                                               |
| D15    | the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing                                                       |
| D19    | "according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."                                                                                                                                                                                                                                                                   |
| D47    | the deed was done by a subject race — by black men who dared to shed the blood of their masters ... British journalist William Howard Russell — 1857 mutiny through British colonial journalist's lens — if Indian content appears at all in MMLU, it is through a colonial/Western perspective                                                                                                                      |
| WEB-15 | ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)                                                                                                                                                                                                                                                                   |
| WEB-13 | MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)                                                                                                                                                                                                                                                    |

## Information Gaps

- Annotator demographics, IAA protocols, label-disagreement aggregation — none documented in paper.
- Whether any cross-cultural annotation comparison has been performed for MMLU labels.

### Requires Expert Verification

- Indian SME re-annotation of moral\_scenarios, jurisprudence, and any Indian-content items to quantify label-flip rate.
- Comparison of MMLU answer keys against UPSC/SSC official answer keys for any overlapping content.

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### Remediation

**Gap 1: US-anchored annotator pool and US-norm scoring (especially moral\_scenarios, jurisprudence, world\_history); zero documented Indian SME review.**

**Recommendation:** Conduct Indian-SME re-annotation of any MMLU items retained for STEM proxy use; for India-relevant content, rely on benchmarks with documented Indian annotation chains (ParamBench [WEB-15], MILU [WEB-13]).

| ID     | Evidence                                                                                                                                                          |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-15 | ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)                |
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